

# Sugarcane & the Land in Brazil

An environmental deep-dive — responding to Paloma's concern about sustainability

Prepared by Sophia Truesight (TrueSight DAO Autopilot) · July 2026 · Sources: MDPI Land, EPA/HERO review, Springer (Agron. Sustain. Dev.), CEPAL/USP, Biofuelwatch, Guardian, Bonsucro, ANP

## 1. Executive Summary

Paloma's concern is fair and worth taking seriously. The honest answer is **nuanced**: Brazilian sugarcane has a genuine sustainability story (decoupled from Amazon deforestation, net carbon removals, falling water intensity, pre-harvest burning phase-out) — but it also has real environmental costs (monoculture pressure, agrochemicals, soil compaction, Cerrado expansion risk, and indirect land-use effects). A responsible buyer like TrueTech Inc. can and should select mills that are certified (RenovaBio / Bonsucro) and verifiably not driving native-vegetation conversion.

**Mission tie-in:** The DAO's mission is restoring **10,000 hectares of Amazon rainforest**. Any sugar sourcing decision should favor suppliers whose operations are **deforestation-free, certified, and land-positive** — so our trade volume reinforces, rather than contradicts, the mission.

## 2. Land-Use Facts: What the Data Actually Shows

### 2.1 Sugarcane expansion is large but overwhelmingly on existing farmland

- Sugarcane expanded **+6.06 million hectares (Mha)** across Brazil's center-south and north between 2000 and 2020, reaching **8.45 Mha** in 2020 (MDPI Land, 2023).
- **98.4% of that expansion was over existing agricultural areas** — mostly poor-quality pasture (55%), cropland (15%), and mosaic land (14%).
- Net result 2000–2020: a **net carbon removal of ~9.8 Tg CO<sub>2</sub>/yr**, driven by converting degraded pasture to cane plus the shift from burned to unburned harvest.
- Why? Sugarcane is **highly productive per hectare** — it produces more energy and sugar per unit of land than most alternatives, so it relieves land pressure rather than adding to it when sited on degraded pasture.

### 2.2 The Amazon: legally protected, and industry has stayed out

- A **2009 federal decree bans sugarcane production in the Amazon biome**; a 2018 senate attempt to lift the ban was defeated after opposition from UNICA (the mills' own union), Greenpeace, and five former environment ministers.
- Amazonian states produced **<1%** of Brazil's cane in 2018/19.
- **RenovaBio requires zero deforestation in sugarcane areas post-2018** — a hard national standard for certified producers.

## 2.3 The real risk zones: Cerrado & indirect effects

- The **Cerrado (savanna) biome** is the actual expansion frontier — pressure there is real, and sugarcane joins soy and pasture as drivers of native-vegetation conversion.
- **Indirect land-use change (ILUC)**: a 2017 study (Jusys) found sugarcane *indirectly* contributed to deforestation 2002–2012 by displacing pasture/crops into forest frontiers. CEPAL/USP modeling shows each extra sugarcane hectare was associated with a 0.47 ha fall in pasture — which must go somewhere.
- **Para note**: ESALQ estimates Para could theoretically host up to 9 Mha of cane — but CONAB puts actual Para harvest at ~736,000 tons, a rounding error nationally. A "Para mill" would be unusual and worth extra due diligence specifically because the region is not a traditional cane belt.

## 3. Water, Soil & Pollution

| Issue                      | Finding                                                                                                                                                                                                                           | Severity / Trend               |
|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|
| <b>Water use</b>           | Brazilian cane is mostly rain-fed; irrigation is limited to specific regions. Bonsucro-certified mills cut water-use costs ~20%.                                                                                                  | Low-to-moderate; improving     |
| <b>Soil erosion</b>        | Ranges 0.1–109 t/ha/yr depending on slope, rainfall, management. Lower than row crops like corn/soy on average, but can be severe on steep or poorly managed land.                                                                | Variable; management-dependent |
| <b>Soil compaction</b>     | Identified as a main issue in sugarcane systems — heavy machinery during planting/harvest.                                                                                                                                        | Persistent                     |
| <b>Agrochemicals</b>       | Monoculture raises pest pressure → higher pesticide use. Localized human-health and water concerns, incl. aerial application near communities.                                                                                    | Real; site-specific            |
| <b>Vinasse</b>             | 7–13 L of vinasse per L of ethanol produced. Historically a pollution hazard (water contamination, odors); today mostly recycled to fields as fertilizer — long-term studies show net soil-health benefits when properly managed. | Managed; improving             |
| <b>Pre-harvest burning</b> | Legally being phased out (Sao Paulo led the way). Unburned (green) harvest improves air quality and adds soil carbon; the burned-to-green transition alone delivered ~16% of the sector's carbon removals.                        | Declining rapidly              |

## 4. Certification & Policy: The Buyer's Toolkit

- **RenovaBio (ANP, 2017– )**: national biofuel certification; carbon-intensity scoring via RenovaCalc; zero-deforestation rule; generates tradable CBIO decarbonization credits. Mills self-report and are audited by accredited inspection bodies.
- **Bonsucro**: global gold-standard sustainability certification for sugarcane — covers GHG, water, biodiversity, labor, living wage. Certified producers show **-31% farm-level** and **-20%**

**mill-level GHG** after 4 years. A 2024 ANP–Bonsucro agreement is integrating the two schemes; a joint Operator's Guide was released June 2026.

- **Fuel of the Future Law (2024/25)**: raises the ethanol blend ceiling, expanding demand — increases both opportunity and scrutiny.
- **EU–Mercosur**: the EU's sustainability criteria for bioethanol hinge on deforestation-free sourcing; a mill that can't prove it risks losing premium EU/China market access. This is now a commercial, not just ethical, driver.

**Key buyer takeaway:** Sustainability is now **tradable and verifiable**. Mills with RenovaBio + Bonsucro certification can prove deforestation-free, lower-carbon production. Mills without it are a reputational and regulatory risk — and increasingly locked out of premium markets.

## 5. What This Means for the 50,000 t/Month Deal

- **Ask for certification up front.** Shortlist mills with RenovaBio (mandatory for ethanol co-producers) and Bonsucro where possible.
- **Add a sustainability clause to the LOI/contract:** no native-vegetation conversion on supply areas; compliance with the 2009 Amazon biome decree; green (unburned) harvest preferred.
- **Due diligence on any "Para mill"** — Para is outside the traditional cane belt and inside the Amazon biome boundary; verify land tenure, CAR registration, and any deforestation linkage before contracting. This connects directly to Milena's point about visiting and certifying facilities.
- **Position it as a feature:** Brazilian cane sugar (esp. certified, low-ILUC) is among the most land-efficient sweeteners globally — ~9.8 Tg CO<sub>2</sub>/yr net removals across the sector. TrueTech can present this to Chinese buyers as a defensible, ESG-clean supply chain.
- **Volumetric context:** 50,000 t/month ≈ 600,000 t/yr — roughly **0.7% of Brazil's ~85 Mt sugar output**. One large certified mill (e.g. Raizen, Tereos, Sao Martinho) can supply it without any land-use change at the margin; it is not a driver of new conversion.

## 6. Balanced Verdict (for the record)

| Paloma's concern                    | Data response                                                                                                                                                                                                                                               |
|-------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| "Sugarcane destroys the Amazon"     | Largely <b>not supported</b> for cane itself: Amazon biome ban since 2009, <1% of production there, 98.4% of expansion on existing farmland. Indirect displacement effects are real but modest (ILUC ~0.08–0.14).                                           |
| "It's not environmentally friendly" | <b>Partial truth:</b> monoculture, agrochemicals, soil compaction, Cerrado pressure, and vinasse risk are genuine. But the sector is among the most regulated and certifiable crops in Brazil, with measurable net carbon removals and a burning phase-out. |
| "We shouldn't do the deal"          | <b>Depends on supplier choice.</b> With certified mills + deforestation-free clauses, the deal can be consistent with the DAO mission. Without them, it can't.                                                                                              |

Prepared for internal TrueSight DAO / TrueTech Inc. discussion. Sources: MDPI Land 12(3):584 (2023); EPA HERO review (Bordonal et al., Agron. Sustain. Dev. 2018); CEPAL/USP ILUC modeling (Bento et al.); Jusys 2017 (Taylor & Francis);

Biofuelwatch (Mendonca); The Guardian (2018); Bonsucro; ANP; USDA FAS; CONAB; MapBiomass. Figures reflect best available public data as of July 2026.